

BAAZ JKBOSE



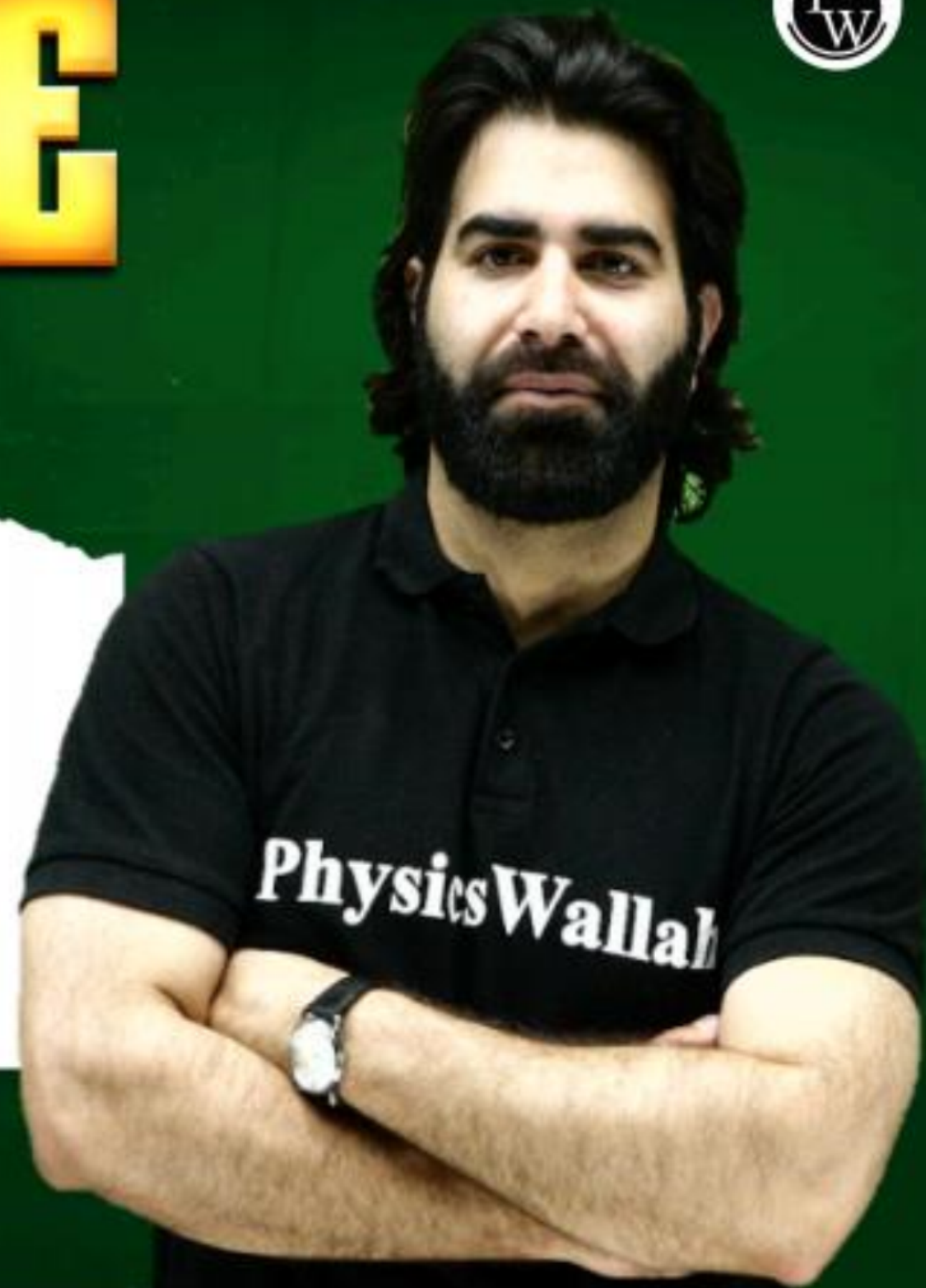
2026 CLASS-11TH

Excretory Products and
Their Elimination

LECTURE-4

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Counter Current Mechanism
Regulation, Hemodialysis

Revision

Urine Formation

$$\text{NFP} = \underline{10 \text{ mmHg}}.$$

$$\text{GFR / Min} = \underline{125 \text{ ml}}.$$

$$\text{Max Reabsorption} = \underline{\text{PCT}}.$$

$$\text{Descending L.O.H} = \underline{\text{Permeable to H}_2\text{O}}.$$

Mechanism of Concentration of Urine

↳ Mammals produce Concentrated Urine [Osmolarity = 1200 mOsm/L]
[4 times than Blood]

↳ For it there should be High Osmolarity of Medullary Interstitium

↳ Medullary Osmolarity remains high Due to

④ Urea Recirculation → Recirculation of Urea Between Collecting Duct & Ascending of Loop of Henle.
→ This increase Amount of Urea in Medulla.
→ Helping in Increasing Osmolarity

⑤ Counter Current Mechanism

↳ Due to this NaCl remains in Medulla.

Cortex. (300)

Outer Medulla (300)

(Urea + NaCl)

Inner Medulla (1200)

⑤ Counter Current Mechanism

↳ This Mechanism is between Loop of Henle & Vasa Recta.

↳ Counter current Means flow in opposite Direction.

↳ This is seen in Loop of Henle & Vasa Recta.



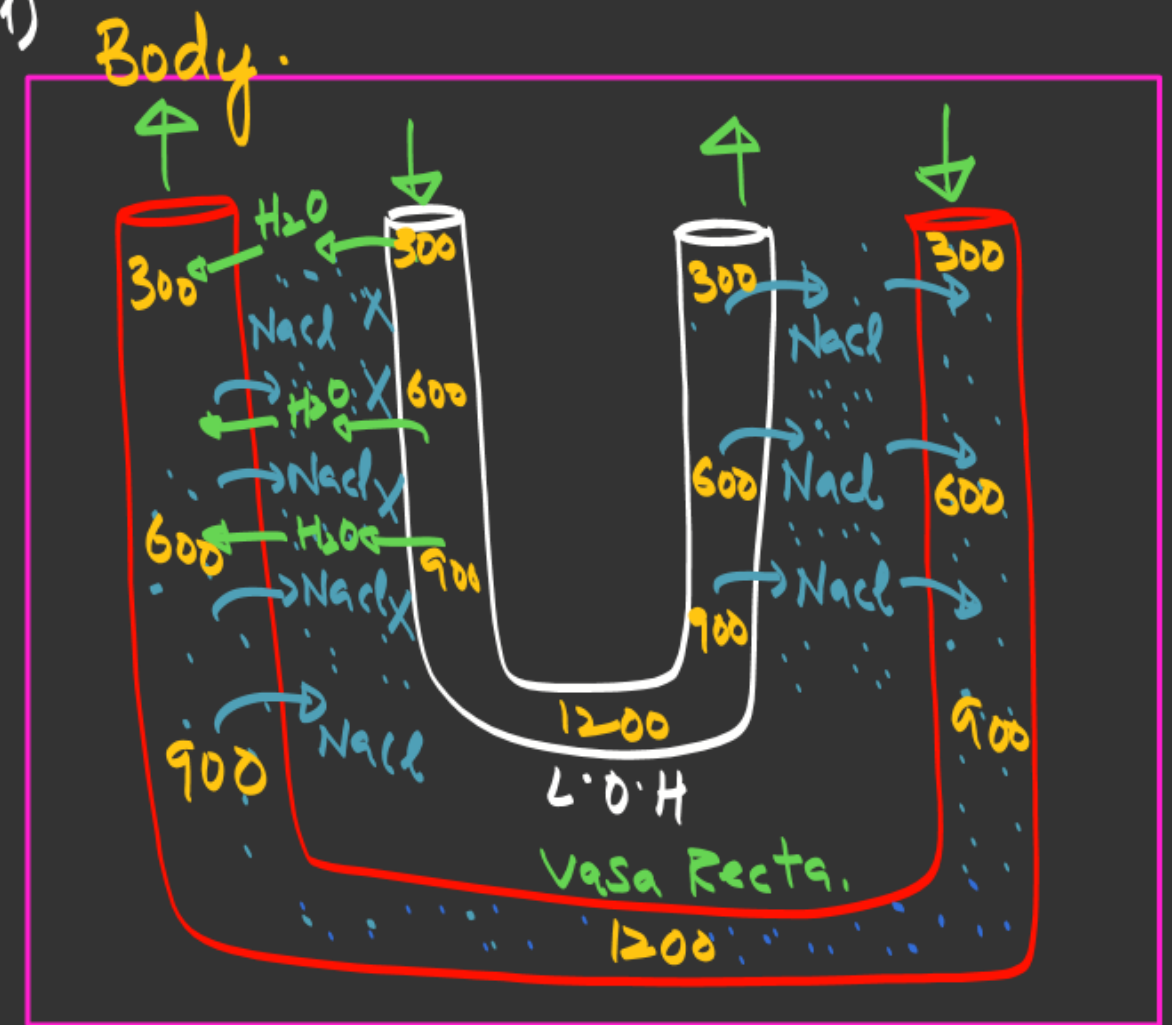
↳ In this Mechanism, Ascending of L.O.H reabsorbs Salt (NaCl) and that is picked by Descending of Vasa Recta.

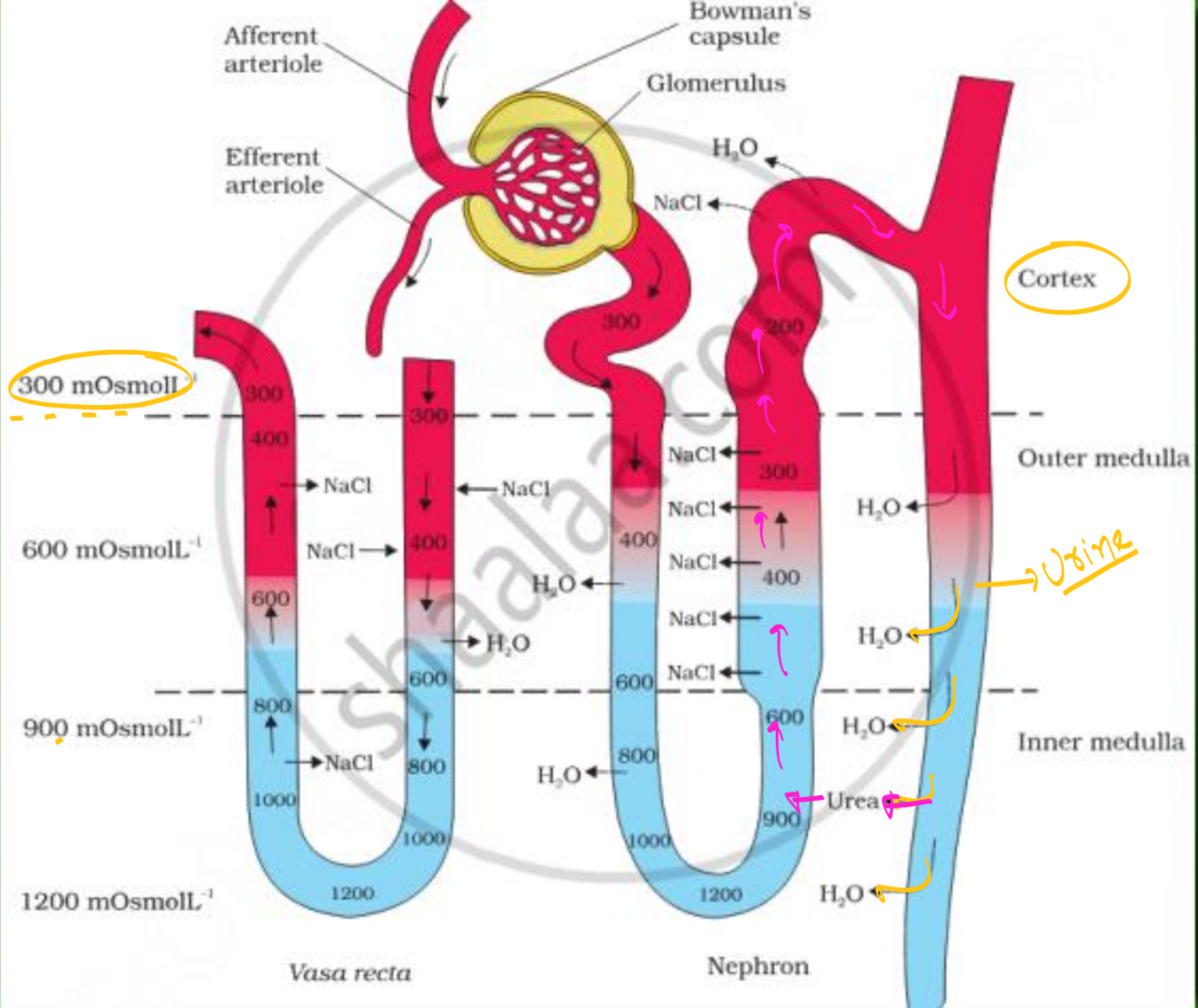
↳ This Salt (NaCl) is given Back to Medulla by Ascending of Vasa Recta.

↳ This Salts Remains in Medulla as Descending of L.O.H is impermeable Salts.

↳ Meanwhile, H₂O from descending Loop Henle goes into Ascending of Vasa Recta.

- Hence, Water goes to Body and Salt goes to Medulla.





Regulation of kidney function / Urine Formation

→ Harmonal Regulation is by three ways;

① ADH

② Renin-Angiotensin Mechanism

③ ANF

① ADH - Means Anti Diuretic Hormone

opposite Diuresis

↳ It is a Hormone, which prevent/opposes Diuresis.

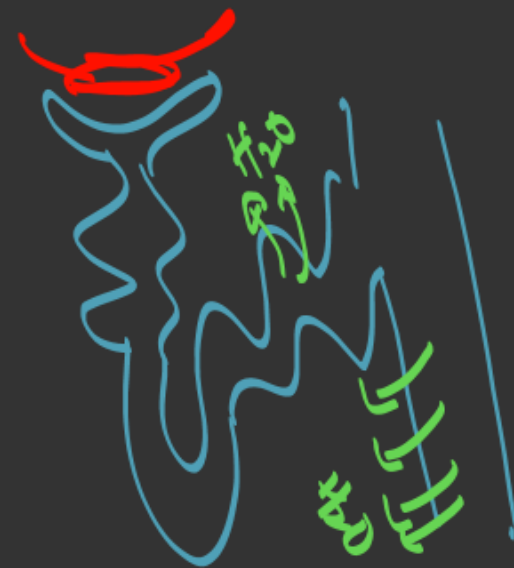
↳ It is released when the Osmolarity of Blood increases ($>300mOsm/L$) or we can say when there is less H_2O in Blood.

- Hormone = chemical Messenger

- Diuresis = formation of Dilute Urine

↳ Osmoreceptors in Hypothalamus sense it, and cause Release of ADH from (Hypothalamus/post Pituitary) (Store)

↳ ADH acts on Later part Tubule and cause H_2O Reabsorption. (DCT & Collecting Duct)



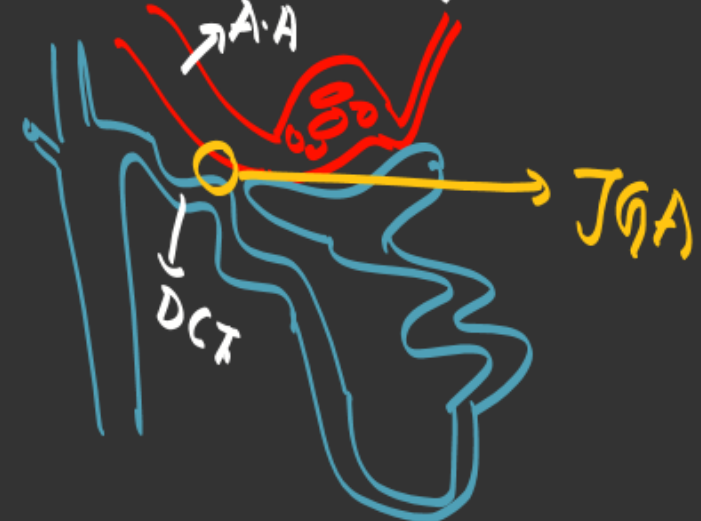
→ ADH is Also vasoconstrictor (↓ Diameter of Blood vessels), causes ↑ in BP.
also increase in GFR, It is also klas Vasopressin

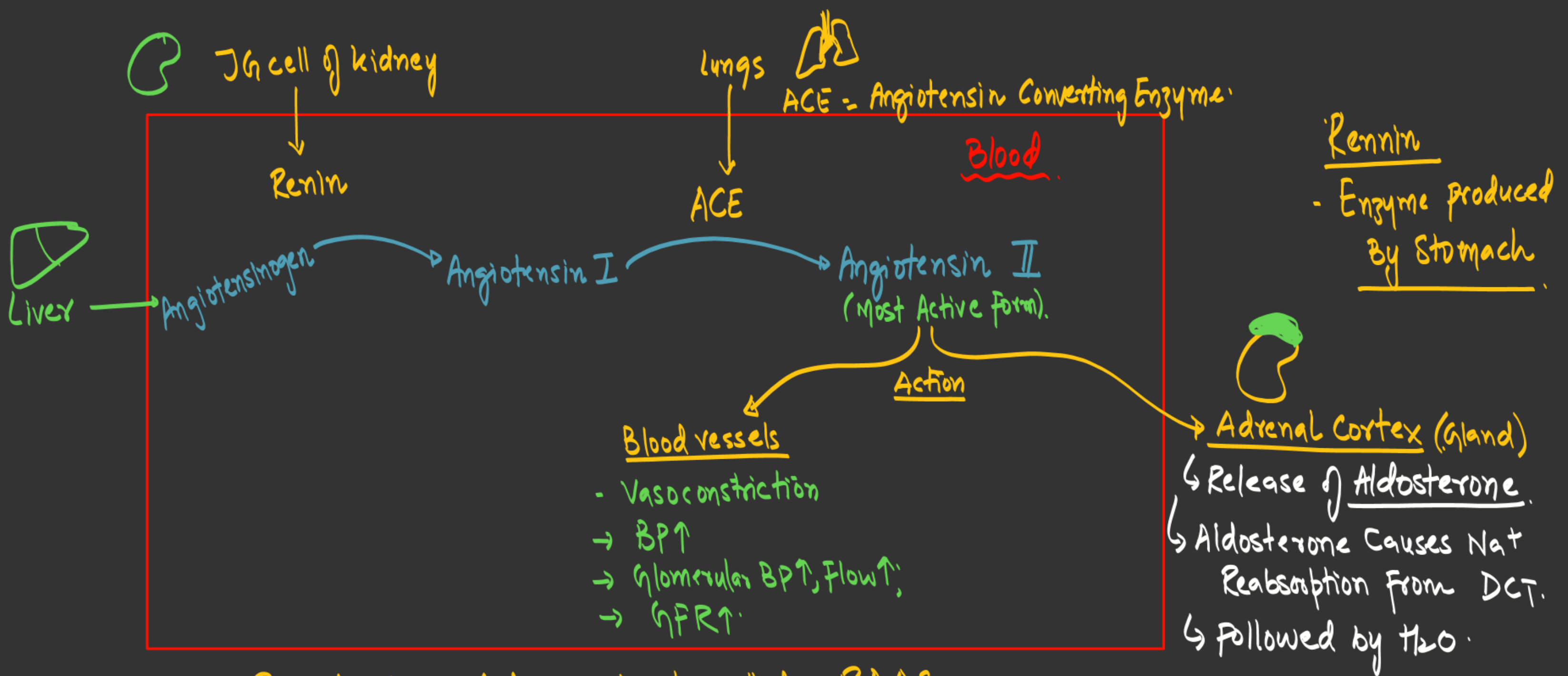
H/W Diabetes Insipidus.

- ① Renin-Angiotensin Mechanism
- Renin is from JGA Apparatus ^{Near} (Juxta Glomerular Apparatus)
 - cells in Afferent Arteriole is called JG-cell.
 - JG-cell Releases Renin, when →
 - ✓ GFR is low, Glomerular BP ↓
 - Glomerular Blood Flow is low
 - ✓ BP ↓, Body Fluid volume ↓
 - ✓ Less Nat in Body.

JG - Apparatus

These are specialised cells in Afferent Arteriole & DCT at the point of their contact





Renin Angiotensin Mechanism is also called RAAS.

Renin Angiotensin Aldosterone System.

Red - presence of RBC / Dark Yellow

③ ANF

→ Atrial Natriuretic Factor

- ↳ It is from Atrium of Heart
- ↳ It is opposite to Renin Angiotensin Mech.
- ↳ It cause Na^+ Excretion.
- ↳ It is vasodilator ($\uparrow$ Blood vessel Diameter)
causing $\downarrow$ in Blood Pressure,
 $\downarrow$ in Glomerular Blood Flow
 $\downarrow$ GFR.



Natutesis

↳ Excretion of $\uparrow$ Amount of Na^+ .

(+/-)

ANF vs RAAS

Thank
You